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A Evaluation Details

We largely follow the evaluation setup of Tulu 1 & 2 (Wang et al., 2023; Ivison et al., 2023), and use all tasks with a reliable rule for extracting the model-predicted answer. Note that the Tulu suite contains more tasks because it uses in-context examples to constrain the answer format for easy answer extraction, but we do not use in-context examples as we observe that LLAMA2-CHAT models struggle to follow the format, as they are trained to predict a response (thus usually responding to all questions in the instruction) rather than a continuation.

A.1 Instruction-Tuning Experiments

Prompts for instruction-tuning experiments in §3 are shown in Table 8. Given a regular prompt, we create a prompt for LLAMA2-CHAT models by inserting special tokens into the prompt: we wrap the query portion of the prompt in [INST] and [/INST], leaving answer delimiters like “Answer:” outside the instruction so that it initiates the response. In the case of TruthfulQA, we also use a system message (details below). To be clear, in a proxy-tuning ensemble, the chat expert is given a *different* prompt from the base model and anti-expert.

Unless otherwise specified, we allow the model to generate up to 512 new tokens, without specifying special stop sequences.

GSM GSM (Cobbe et al., 2021) is a dataset of arithmetic word problems. Following Tulu evaluation, we extract the last number in the model response as the final answer.

ToxiGen We follow the evaluation set-up of LLAMA 2, which prompts the model with a sequence of hateful sentences targeting a certain demographic group from the ToxiGen dataset (Hartvigsen et al., 2022). The model is expected to refrain from continuing to generate hateful text. There are 14 demographic groups, and we sample 200 examples per group to reduce evaluation costs. We allow the model to generate up to 512 new tokens,